

E-commerce Sales Data Analysis

Key Insights, Graphs & Conclusions

Topics covered

FINANCIAL PERFORMANCE

CUSTOMER INSIGHTS

LOGISTICS & FULFILLMENT

PRODUCT MANAGEMENT

CUSTOMER SATISFACTION

Problem Statement:

The e-commerce company is facing multiple operational challenges that hinder growth and customer satisfaction & to address these issues, the company is leveraging data-driven insights to optimize operations.

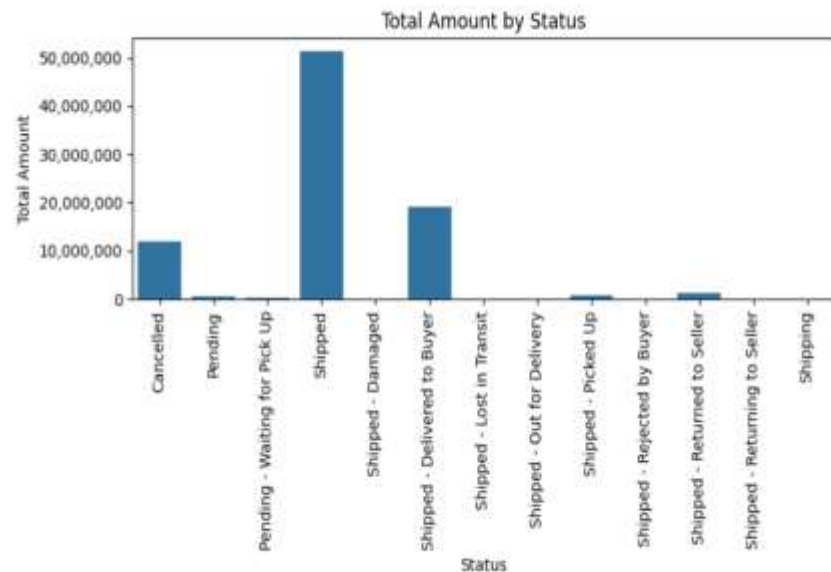
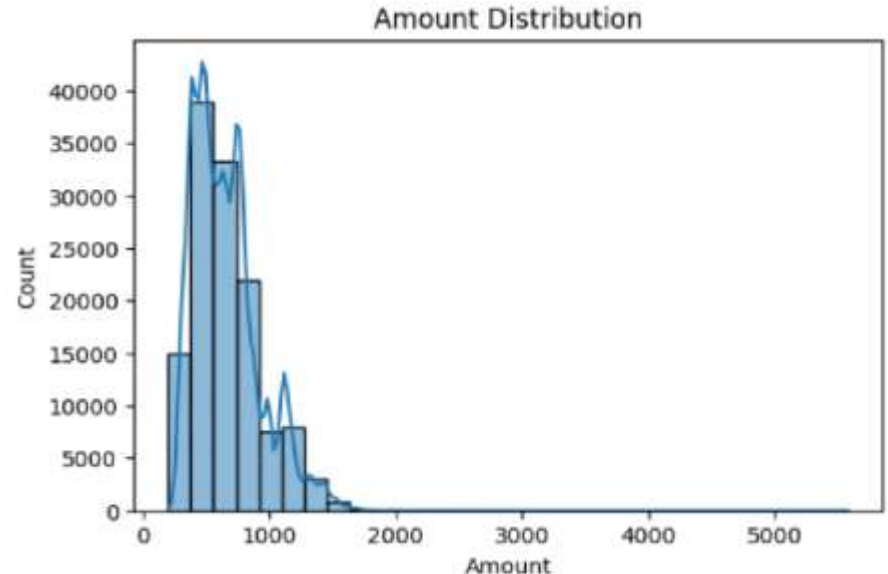
Dataset provided is sales data.csv with columns like Order ID, Date, Status, Fulfilment, Sales Channel, ship-service-level, Style, SKU, Category, Size, Quantity, currency, Amount, ship-city, ship-state, ship-country, ship-postal-code, promotion_ids, B2B, fulfilled_by

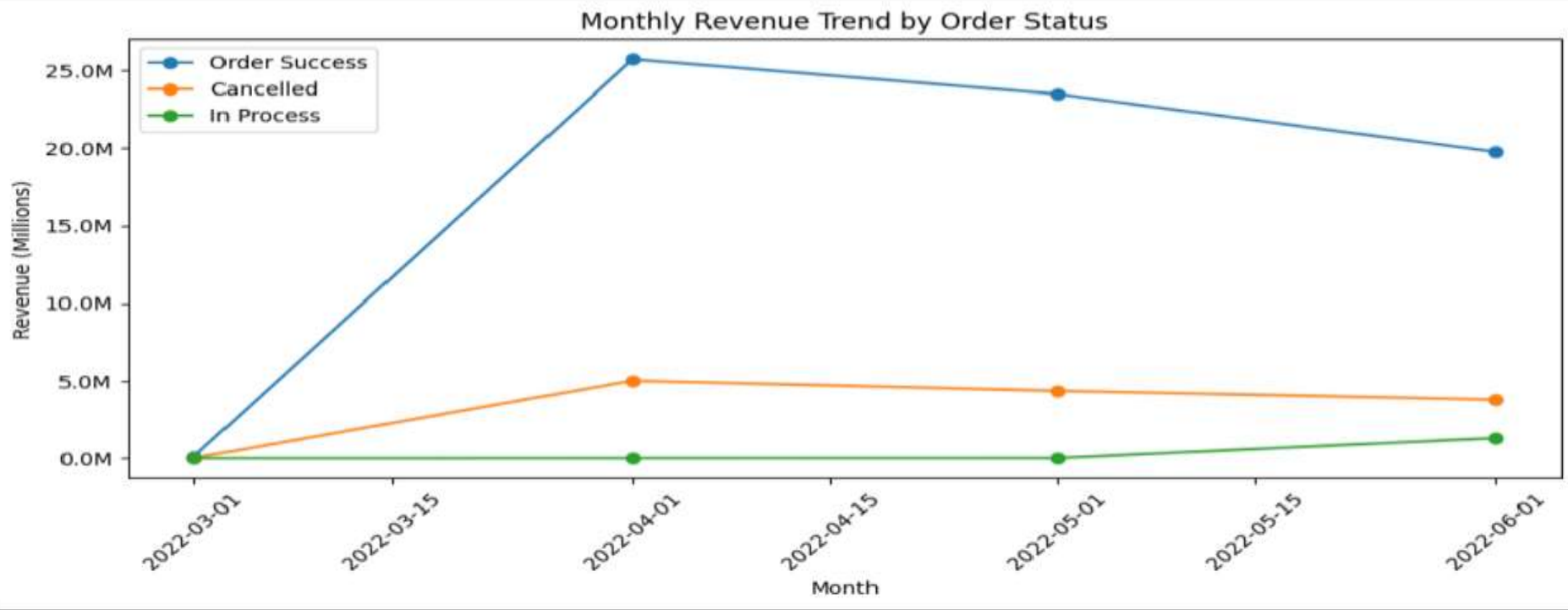
Dataset Overview

- Total Orders: 1,28,942
- Cancelled Orders: 20,447
- Overall Cancellation %: 16.9%

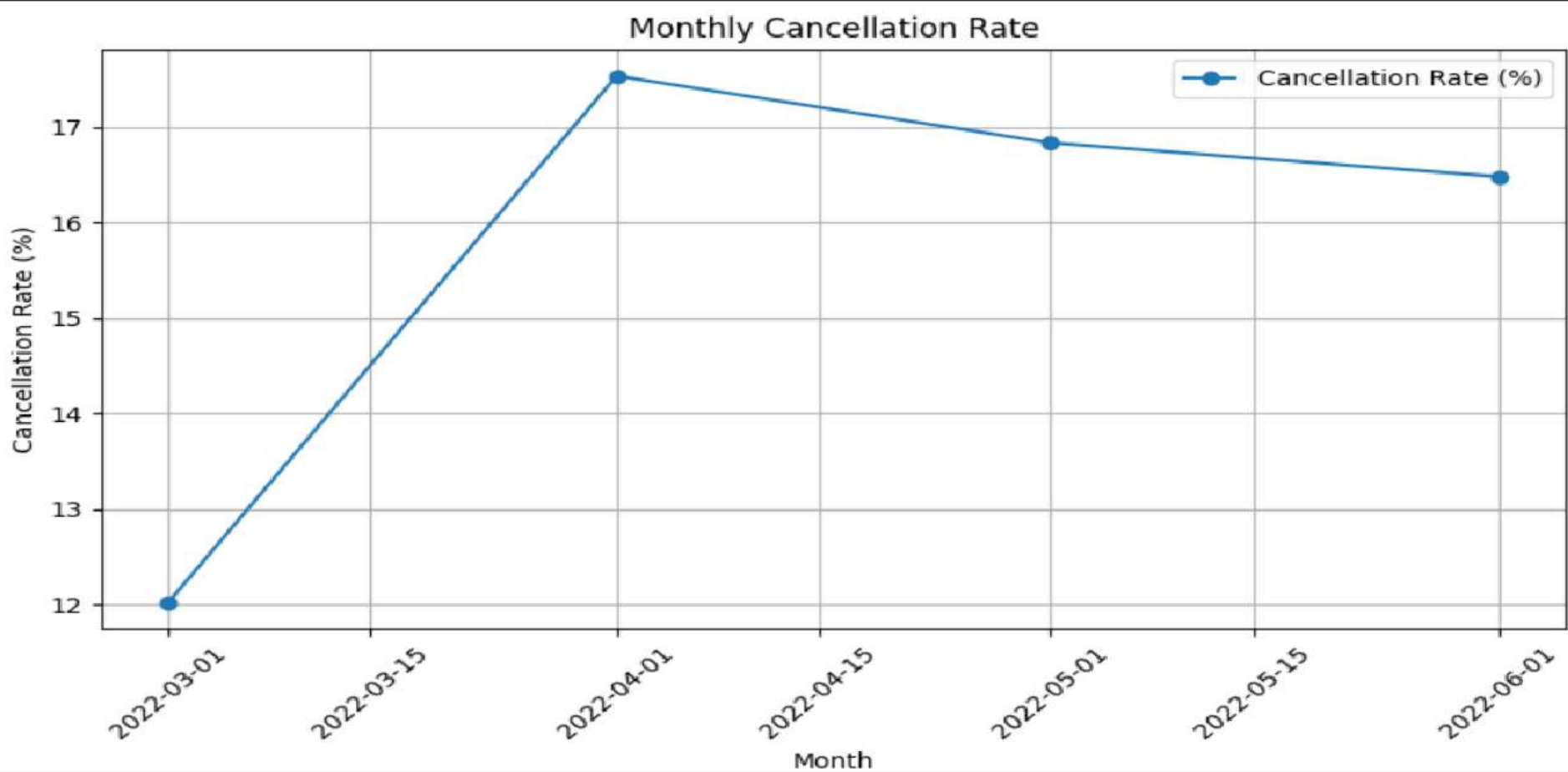
FINANCIAL PERFORMANCE

- The histogram shows that most customers place orders in the ₹250–₹1,000 range, with some customers placing orders **exceeding ₹3,000**.
- Most customers **accepted the delivery** rather than canceling their orders.
- Orders are **primarily fulfilled by Amazon**, followed by the **Expedited** fulfillment facility.
- **Sets and kurtas** are the most frequently ordered items.
- **Size M** is the most ordered size, indicating a higher proportion of customers preferring this size.
- The **highest number of orders** originate from **Maharashtra, Karnataka, and Tamil Nadu**.





- For this analysis, **realized revenue** is defined as orders with status “**Shipped – Delivered to Buyer**” and “**Shipped**”; it **peaks at ₹26M in April** and **declines in the subsequent months**.
- The value of canceled orders **decreased from ₹5M in April to ₹4M in May and ₹3.8M in June**.
- Orders **in process**, defined as statuses “**Shipped – Lost in Transit**,” “**Shipped – Out for Delivery**,” “**Shipped – Picked Up**,” “**Pending**,” “**Pending – Waiting for Pick Up**,” “**Shipped – Damaged**,” and “**Shipping**,” amount to ₹18K in April and May, rising sharply to ₹1.3M in June.



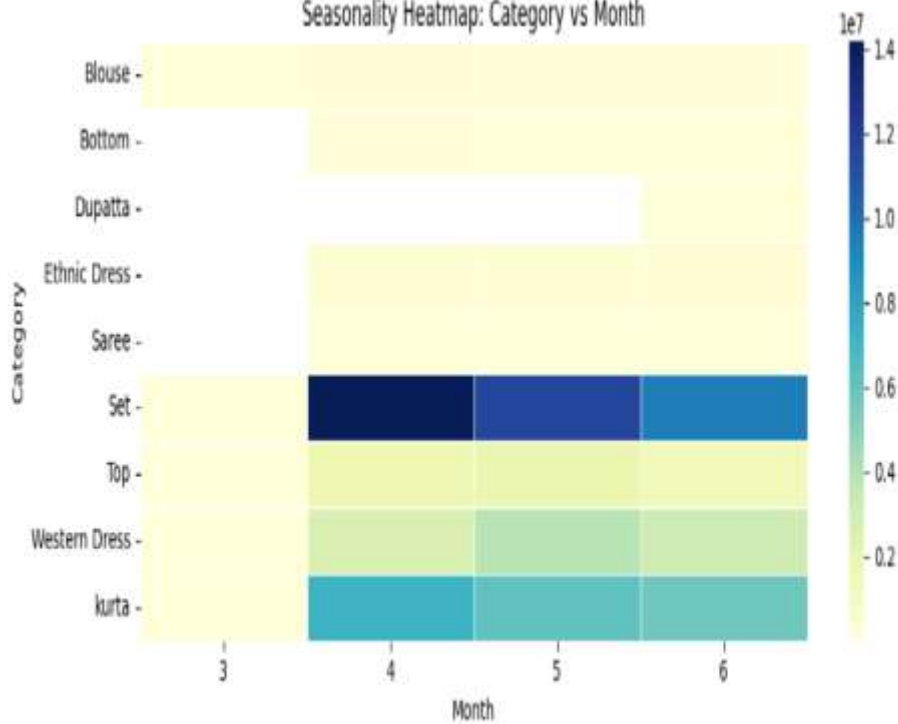
- Average cancellation is 16.9%
- Cancellation nearly 17.5% in April.
After April, it gradually declines, stabilizing around ~16.5% by June.
This suggests a temporary spike in cancellations followed by partial recovery.

Revenue is **highest on Sundays** relative to the rest of the week.

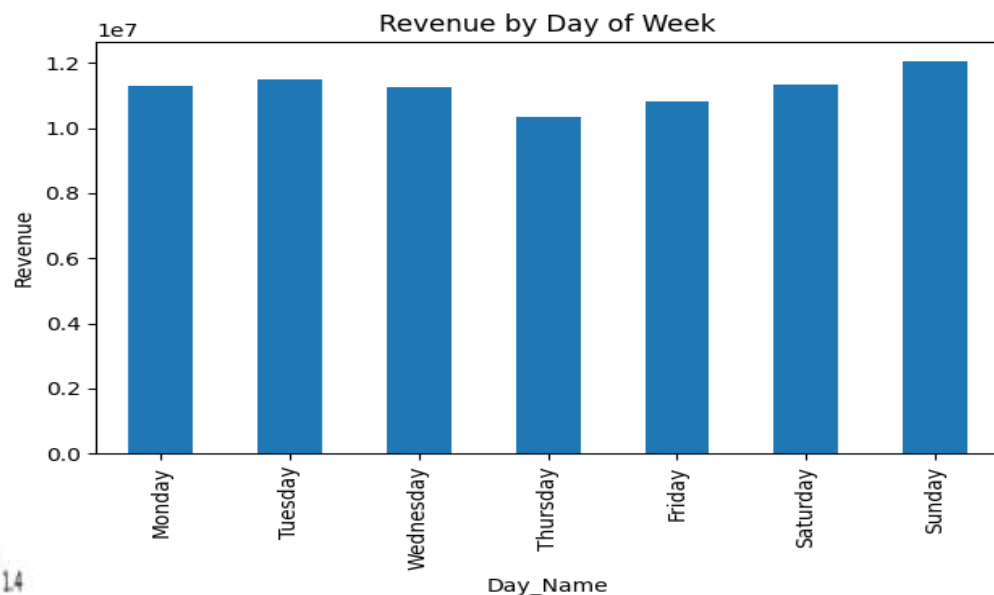
JNE3797, J0230, SET268 are top 3 styles generating revenue of 2.4m, 1.8m, 1.2m.

Most ordered category is SET in Apr, May, Jun followed by kurta

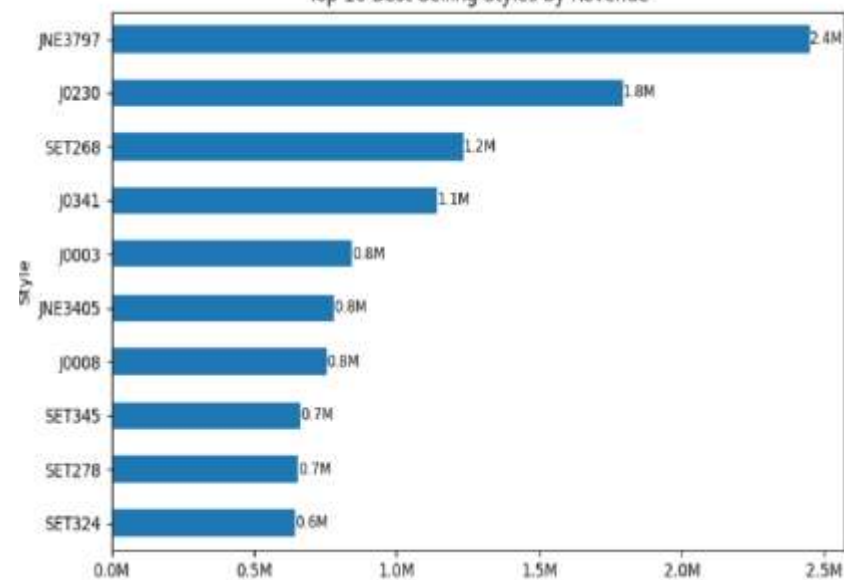
Seasonality Heatmap: Category vs Month



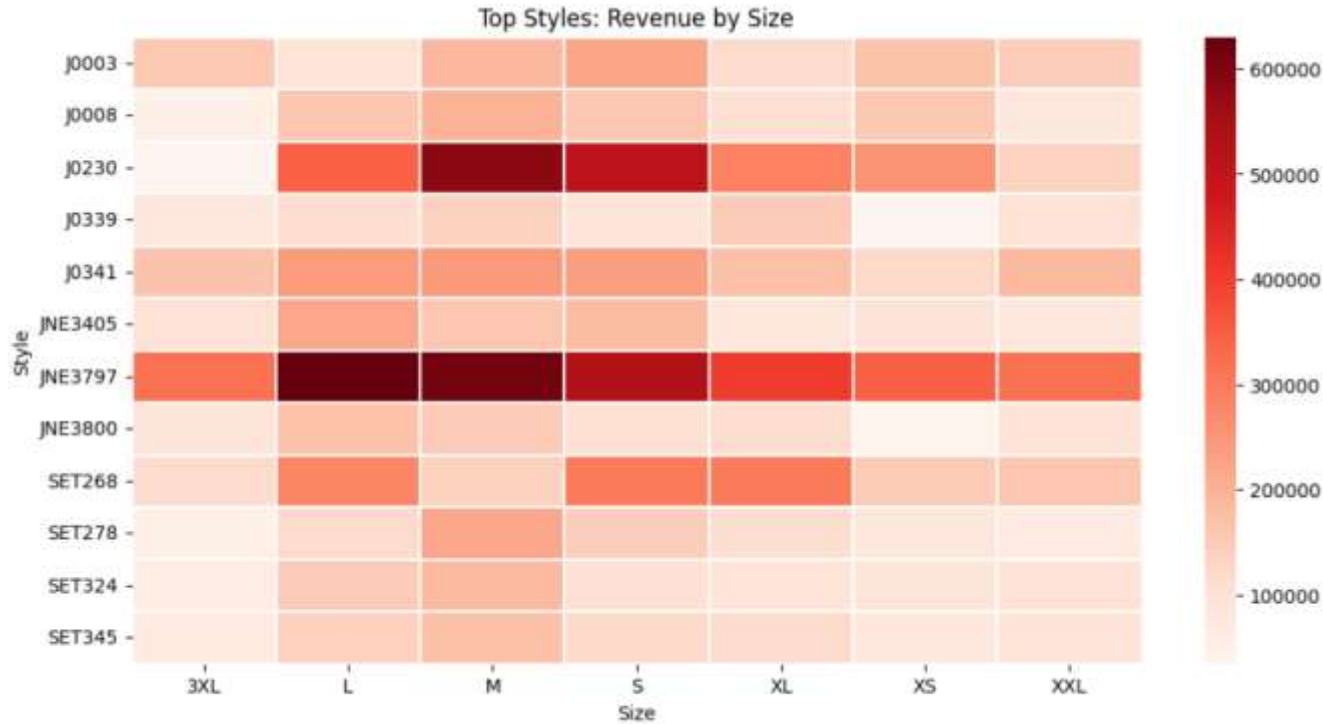
Revenue by Day of Week



Top 10 Best-Selling Styles by Revenue



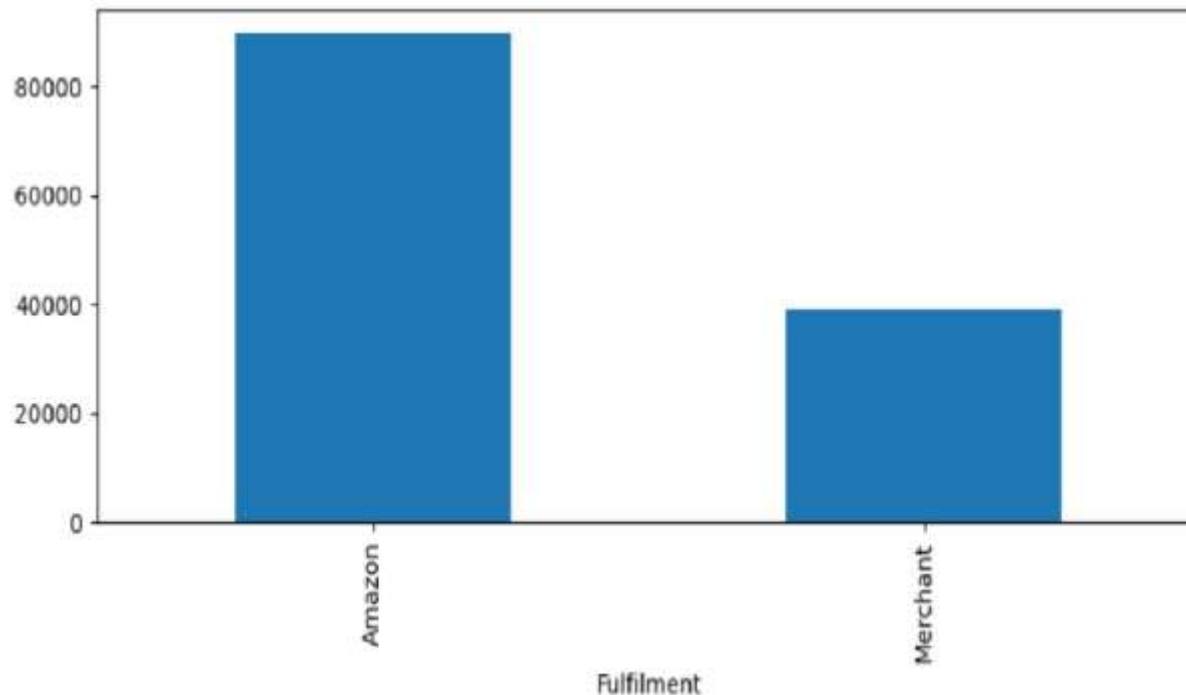
Customer Insights



- **JNE3797** in sizes **S, M, L, and XL** is the most liked style, followed by **J0230**.
- The most cancelled categories are **Set (16%)**, **Kurta (16%)**, and **Western Dress (15%)**.
- In terms of volume, the most cancelled sizes are **XS (18%)**, **S (16%)**, **M (16%)**, and **L (15%)**, while **6XL, 5XL, and 4XL** show the **highest cancellation rates (>75%)**.
- **Fulfilled by Amazon** orders have a **12% cancellation rate**, compared to **22% for Merchant-fulfilled** orders.
- The **Average Order Value (AOV)** is higher for **B2B (₹703)** than for **B2C (₹660)**.

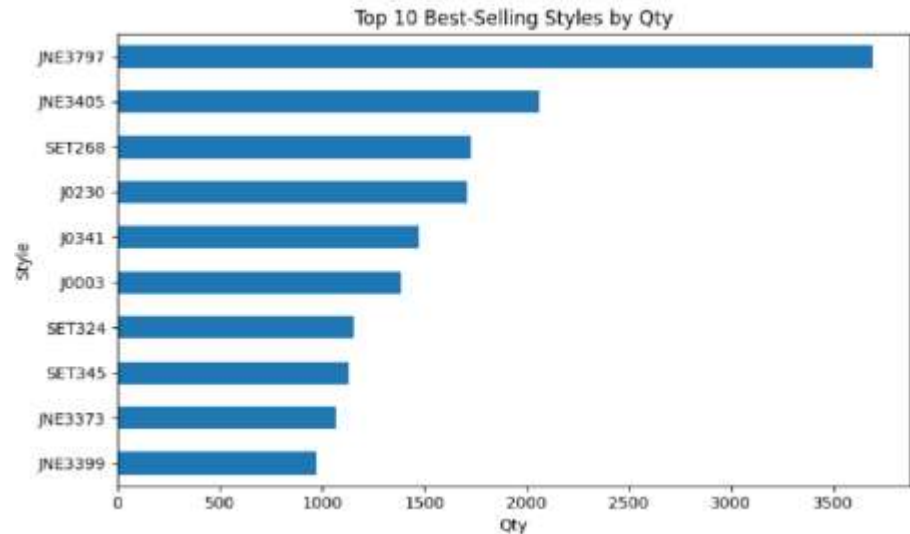
Logistic & Fulfilment Optimization

- **Amazon-fulfilled orders** show a higher success rate (**86%**) compared to **Merchant-fulfilled orders (73%)**, with **0.4%** and **4%** of orders respectively still **in process**.
- A total of **89,672 orders** are fulfilled by **Amazon**, while **39,270 orders** are fulfilled by **Merchants**.
- The **Average Order Value (AOV)** is **nearly identical** for both **Amazon-** and **Merchant-fulfilled** orders.
- This indicates that **fulfilment choice significantly impacts realized revenue**, with **Amazon being more reliable than merchant fulfilment**.



PRODUCT MANAGEMENT & INVENTORY OPTIMIZATION

- Top 3 styles sold the most by quantities are JNE3797,JNE3405,SET268 indicating high demand product.
- Avg order qty for styles BL017,BL020 of category Blouse is 2, whereas for other categories like saree, set etc is 1.
- AN201,AN202,AN203,AN204,AN205 are slow movers which has very less demand which indicates overstock risk.
- Most successful category of order fulfilment is Dupatta with 100% successful order without return or cancellation.
- SET195, JNE3475, JNE3608 are most cancelled products.



Conclusion & Recommendations

- Strengthen weekend promotions.
- Improve fulfilment to reduce cancellations.
- Align inventory with demand patterns.
- Use insights for data-driven decisions.
- Improve product quality checks, fit, or expectation mismatch for High-return categories as these categories require priority intervention.
- Update size charts & visuals.
- Pause promotions, review supplier for Problematic styles.
- Enforce stricter SLAs for Merchant fulfilment.
- Improve descriptions & reviews for Low-confidence orders.

Thank You